Name ____ Date ___________ Period _____

Parent Functions I: √x and |x|

TEACHER KEY | Algebra II | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph the square root and absolute value parent functions from a table and analyze their key attributes: domain, range, intercepts, symmetry, and minimum.

Vocabulary

Reflectional symmetry — the graph can be **folded** across a line and land exactly on itself.

x-intercept(s) — where the graph crosses the x-axis. There, y = **0**. **y-intercept(s)** — there, x = **0**.

Zeros — the x-value(s) that make a function equal **0** — the same locations as the x-intercepts.

Part 1 — The Square Root Function, f(x) = √x

Fill in the table. Watch what happens with negative inputs.

x	-4	-1	0	1	4	9
f(x) = √x	undef.	undef.	0	1	2	3

Why are the first two undefined? **you cannot take the square root of a negative number**

Part 2 — The Absolute Value Function, f(x) = |x|

Absolute value gives distance from zero, so the output is never negative.

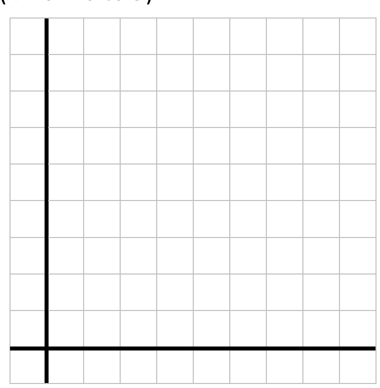
x	-5	-2	-1	0	1	2	5
f(x) = |x|	5	2	1	0	1	2	5

What shape does this make? **a V with its vertex at the origin**

Part 3 — Graph Them Both

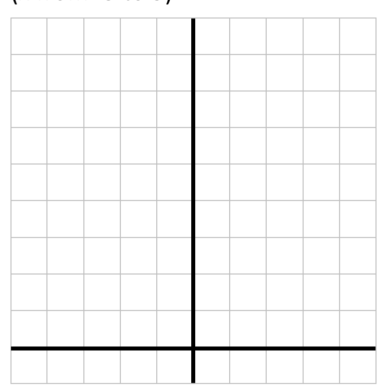
Each square is 1 unit. Plot your table values, then connect them smoothly.

A. f(x) = √x (x from 0 to 9)



Which quadrant only? **I**

B. f(x) = |x| (x from -5 to 5)



Fold line? **the y-axis**

Part 4 — Compare the Key Attributes

Attribute	**f(x) = √x**	**f(x) = |x|**
Domain (interval)	**[0, ∞)**	**(-∞, ∞)**

Range (interval)	**[0, ∞)**	**[0, ∞)**
x- and y-intercept	**(0, 0)**	**(0, 0)**
Symmetry	**none**	**across the y-axis**
Minimum	**y = 0**	**y = 0**
Increasing / decreasing	**always increasing**	**down then up**

Biggest difference: their **domains**. Only **√x** refuses negative inputs.

Part 5 — Name That Function

Use only the attributes given. No equation.

1. Domain and range both [0, ∞), no symmetry, always increasing. **f(x) = √x**
2. Domain all reals, range [0, ∞), symmetric across the y-axis, minimum 0. **f(x) = |x|**

Practice

Evaluate, then answer. Show your work where there is room.

1. Evaluate: √25 = **5**   √0 = **0** |-7| = **7** |0| = **0**

2. Explain why √-9 is not a real number.

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3. Name every x-value where |x| = 4. **x = 4 and x = -4** How many? **2**

4. Name every x-value where √x = 4. **x = 16 only** How many? **1**

5. Items 3 and 4 have different numbers of answers. What does that tell you about the two graphs?

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6. On [1, 4], give the minimum and maximum of f(x) = √x. min **1** max **2**

Exit Ticket

Both functions have a minimum of 0 and pass through the origin. Give one attribute that tells them apart, and explain why it happens.

Attribute: **the domain**

Why: **|x| accepts every real number, but √x needs x ≥ 0**